

WEEK 12 ULTIMATE MASTER GUIDE

NPTEL: Introduction to Environmental Engineering & Science

Exam Target: April 17, 2026 | **Focus:** Cape Town Water Crisis, Panaji SWM, Bhopal, Food Waste, Marine Debris & Global Disasters

SECTION 1: THE FINAL EXAM TRAP LIST

- **The Freshwater Ratio Trap:** Only **2.5%** of global water is fresh. Of that, **~69%** is locked in Glaciers/Ice Caps, and only **~30%** is Groundwater. Glaciers — NOT groundwater — are the major portion.
 - **The Cape Town Usage Trap:** Unlike most of the world where agriculture is #1, in Cape Town **Urban/Industrial use (70%)** far exceeds Agriculture (30%). Do NOT reverse these.
 - **The Marine Plastic Trap:** **90%** of floating marine debris is plastic. **62%** of that plastic consists specifically of **Food and Beverage packaging**.
 - **The Mediterranean Trap:** Cape Town's Mediterranean climate means it receives rainfall in **Winter (May–August)** and is dry in Summer.
 - **The GPGP Microplastic Trap:** While the Great Pacific Garbage Patch covers 1.6 million km², **94%** of the estimated 1.8 trillion pieces of plastic in it are specifically **microplastics**.
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SECTION 2: THE CAPE TOWN WATER CRISIS (Running on Empty)

1. “Day Zero”: This does NOT just mean a “water shortage.” Day Zero is the apocalyptic date when the city officially **shuts down the municipal piped water supply**, forcing citizens to line up at public taps for daily rations.

2. The Mediterranean Climate Trap: Cape Town has a Mediterranean climate — it gets its rain in **Winter (May–August)** and is completely dry in Summer. Three consecutive dry winters (2015–2018) caused the crisis.

3. Who Uses the Water? (The 70/30 Rule):

- **70%** Urban and Industrial use
- **30%** Agriculture
- (MCQ Trap: Do NOT reverse these! Agriculture uses less than the city.)

4. Water Demand Management (WDM) — Specific Initiatives (All of the Above MCQ):

Cape Town used all four of these simultaneously:

1. **Restrictions** (mandatory usage limits)
 2. **Tariff hikes** (making water expensive to deter overuse)
 3. **Pressure management** (reducing pipe pressure to stop bursting and reduce flow)
 4. **Smart meters** in schools
 5. **Behavioral nudges** — the “Cape Town Water Map” publicly shamed water wasters
- **Unit Conversion: 1 Gallon = 3.785 Litres** (often rounded to **3.7 L** in MCQs).

SECTION 3: PANAJI’S GOLDEN SWM RULE (The 5 Bins)

Panaji achieved a zero-landfill, 100% door-to-door waste collection system through aggressive source segregation. You MUST know the 5 bin colors and their collection frequencies:

Bin Colour	Waste Type	Collection Frequency
Green	Wet waste (food/organic)	Daily
Black/Grey	Glass and metals	Twice a week
Pink	Paper and cartons	Twice a week
Orange	Plastics	Twice a week
White	Non-recyclables/Diapers	Twice a week

- **Rule of thumb:** Only Green (wet waste) is collected daily. All other bins = twice a week.

SECTION 4: THE BHOPAL GAS TRAGEDY (The Nightmare)

5. The Chemistry:

On **December 3, 1984**, water entered a storage tank at the Union Carbide plant. This caused a massive exothermic reaction, releasing approximately **40 tons of Methyl Isocyanate (MIC)** gas over the city.

Chemical Sequence:

Methylamine + Phosgene MIC + 1-Naphthol Carbaryl (Sevin)

- **The final pesticide product** being manufactured was **Carbaryl**, marketed as **Sevin**.
- **Prime Accused CEO:** Warren Anderson (Union Carbide Corporation).

6. Absolute Liability:

The legal principle established in India after this disaster. A factory dealing with hazardous materials has **absolutely NO excuse** if something goes wrong — they must pay damages regardless of negligence. No exceptions.

SECTION 5: FOOD WASTE & REVOLUTIONARY TECH

7. SecondBite (Australia):

An organization tackling food waste through **redistribution** — not composting or recycling. They rescue surplus food from farmers and supermarkets and give it directly to community groups.

8. Hydrothermal Carbonization (HTC — “Zero Loss”):

Developed by **IIT Kharagpur**, HTC converts wet biomass directly into fuel.

Feature	Detail
Input	Wet biomass (no drying needed)
Output	Hydrochar (Bio-coal)
Temperature	180–250°C
Key advantage	NO drying step required — beats incineration on energy efficiency

9. Global Food Waste Framework:

- **Highest Loss Region: Industrialized Asia** has the highest share (**28%**) of global food loss and waste.

- **Target-Measure-Act Framework:** The “**Measure**” step is critical — it identifies *where* and *how much* food is being lost before action can be taken.

10. Virtual Water:

The volume of freshwater used to produce a product, measured at the place where the product was actually produced.

Product	Virtual Water Footprint
1 cup of coffee	130 L (lowest)
1 kg of sugar	1,800 L
1 kg of beef	~15,000 L (highest)

SECTION 6: MARINE DEBRIS

- **90%** of floating marine debris is plastic.
- **62%** of that plastic is specifically **Food and Beverage packaging**.
- **The Great Pacific Garbage Patch (GPGP):**
 - Size: **1.6 million km²**
 - Contains an estimated **1.8 trillion pieces** of plastic.
 - **94%** of those pieces are **microplastics**.
 - Location: **Central North Pacific Ocean**.

SECTION 7: GLOBAL ENVIRONMENTAL DISASTERS

CHEAT SHEET

Disaster	Location & Year	Key Cause	Key Consequence
Love Canal	Niagara Falls, NY; 1942–1953	Hooker Chemical dumped 21,000 tons of toxic waste	Created CERCLA (Superfund Act) , 1980
Minamata Disease	Japan, 1956	Methylmercury in industrial wastewater from Chisso Corporation	Severe neurological syndrome in thousands

London Smog	London, December 1952	Cold anticyclone trapped coal combustion pollutants	Led to UK's Clean Air Act of 1956
Bhopal Gas Tragedy	Bhopal, India, Dec 3, 1984	40 tons of MIC released from Union Carbide plant	Established Absolute Liability principle in India
Chernobyl	Pripyat, Ukrainian SSR, Apr 26, 1986	Explosion at No. 4 reactor	Worst nuclear disaster in history
Flint Water Crisis	Michigan, USA, 2014	City switched to Flint River; insufficient treatment lead leached from old pipes	100,000+ residents exposed to lead

SECTION 8: THE 100% COMPLETE W12 Q&A BANK

The Cape Town Water Crisis

- **Q:** What is the main cause of the water shortage in Cape Town? [2024, 2025]
 - **A:** Abnormally low rainfall combined with increasing demand.
- **Q:** What does "Day Zero" refer to in the context of the Cape Town water crisis? [Slide Veto]
 - **A:** The date when the municipal piped water supply is completely shut down.
- **Q:** What type of climate does Cape Town have, and when does it receive its rainfall? [Slide Veto]
 - **A:** Mediterranean climate; it receives its rainfall during Winter (May–August).
- **Q:** How is water usage distributed between urban/industrial and agriculture in the Western Cape? [Slide Veto]
 - **A:** 70% Urban and Industrial; 30% Agriculture.
- **Q:** Which was NOT a WDM strategy used in Cape Town? [2024, 2025]
 - **A:** Unrestricted agricultural water use. (Agriculture faced 45% reduction targets.)
- **Q:** What specific WDM initiatives did Cape Town use? (All of the Above MCQ)
 - **A:** Restrictions, tariff hikes, pressure management, and smart meters in schools.
- **Q:** What is the unit conversion for 1 Gallon to Litres?
 - **A:** 1 Gallon = 3.785 L (rounded to 3.7 L in MCQs).
- **Q:** What percentage of global water is freshwater?

- **A:** 2.5%.
- **Q:** Is groundwater the major portion of the freshwater reserve?
 - **A: False.** Glaciers/Ice Caps are the major portion (~69%). Groundwater is ~30%.

Panaji SWM & Sorting

- **Q:** Which bin is used for wet waste in Panaji, and how often is it collected? [Slide Veto]
 - **A:** Green bin; collected **daily**.
- **Q:** Match the Panaji bin colors to their waste streams: Black, Pink, Orange, White. [Slide Veto]
 - **A:** Black = Glass/Metals. Pink = Paper/Cartons. Orange = Plastics. White = Non-recyclables.
- **Q:** How often are the non-green bins collected in Panaji?
 - **A:** Twice a week.

Bhopal Gas Tragedy

- **Q:** Which gas was primarily responsible for the 1984 Bhopal Gas Tragedy? [2024, 2025]
 - **A:** Methyl Isocyanate (MIC).
- **Q:** What was the final pesticide product being manufactured at the Union Carbide plant? [Slide Veto]
 - **A:** Carbaryl (Sevin).
- **Q:** What was the chemical sequence used to produce Carbaryl?
 - **A:** Methylamine + Phosgene MIC + 1-Naphthol Carbaryl.
- **Q:** Approximately how much MIC was released in the Bhopal disaster?
 - **A:** ~40 tons.
- **Q:** Who was the prime accused CEO in the Bhopal Gas Tragedy? [Slide Veto]
 - **A:** Warren Anderson.
- **Q:** Which legal principle was established in India after Bhopal? [Slide Veto]
 - **A:** Absolute Liability — hazardous industries are strictly liable with no exceptions.

Food Waste & Technology

- **Q:** What is the primary operational mechanism of the “SecondBite” organization? [Slide Veto]
 - **A:** Redistribution of surplus food from farmers and retailers to community groups.

- **Q:** What is the primary advantage of HTC over traditional incineration? [2024, 2025]
 - **A:** It can process wet biomass directly without the need for an energy-intensive drying step.
- **Q:** What is the primary solid fuel product generated from HTC? [Slide Veto]
 - **A:** Hydrochar (Bio-coal).
- **Q:** At what temperature range does HTC operate? [Slide Veto]
 - **A:** 180°C to 250°C.
- **Q:** Which Indian institute developed the “Zero Loss” HTC technology? [Slide Veto]
 - **A:** IIT Kharagpur.
- **Q:** Which region has the highest share of global food loss and waste?
 - **A:** Industrialized Asia (28%).
- **Q:** Which food item has the lowest virtual water footprint among: 1 kg sugar, 1 kg beef, 1 cup coffee?
 - **A:** 1 cup of coffee (130 L).
- **Q:** What is “Virtual Water”?
 - **A:** The volume of freshwater used to produce a product, measured at the place of production.

Marine Debris

- **Q:** What percentage of floating marine debris is plastic?
 - **A:** 90%.
- **Q:** What specific type of plastic makes up 62% of marine plastic debris?
 - **A:** Food and Beverage packaging.
- **Q:** The Great Pacific Garbage Patch is located in which body of water?
 - **A:** The central North Pacific Ocean.
- **Q:** What percentage of the 1.8 trillion pieces of plastic in the GPGP are microplastics?
 - **A:** 94%.

Global Environmental Disasters

- **Q:** Which chemical company was responsible for toxic waste dumping at Love Canal?
 - **A:** Hooker Chemical Company.
- **Q:** What US legislation resulted directly from the Love Canal disaster?
 - **A:** CERCLA (The Superfund Act), 1980.
- **Q:** Minamata disease was primarily caused by the industrial release of which toxic substance?
 - **A:** Methylmercury (from Chisso Corporation).

- **Q:** The 1952 London Smog, which led to the 1956 Clean Air Act, was primarily driven by burning which fuel?
 - **A:** Coal.
 - **Q:** What caused the Flint Water Crisis (2014)?
 - **A:** The city switched its water source to the Flint River; insufficient treatment caused lead to leach from old pipes, exposing 100,000+ residents.
 - **Q:** At which reactor did the Chernobyl disaster occur?
 - **A:** No. 4 reactor, near Pripyat, Ukrainian SSR.
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SECTION 9: ACTIVE RECALL & FEYNMAN CHALLENGE

Cover the answers below and test yourself before the exam.

Quiz 1: If 90% of marine debris is plastic, what specific item makes up the majority (62%) of that plastic?

***Answer:** Food and Beverage packaging.*

Quiz 2: Rank these in order of water footprint (highest to lowest): Indonesia, UK, USA, Australia.

***Answer:** USA Australia Indonesia UK.*

Feynman Challenge: Explain the “Absolute Liability” principle established after Bhopal.

***Example Answer:** Normally, if a factory has an accident, they can argue it wasn't their fault (e.g., “it was a natural disaster” or “a worker made a mistake”). **Absolute Liability** says: If you are doing something inherently dangerous (like making toxic MIC gas) and it leaks, you are **100% responsible** for all damage, period. No excuses, no exceptions.*